Аналитика и модель

Индекс → признаки → target → baseline → ограничения

Что нужно знать

- The Champagne Index formula: recorded goals plus assists, divided by the matches the source reports for that club-season. Because goals and assists are counted separately, one goal with an assist contributes twice. It is our own proxy for attacking output, not xG and not an official rating.
- The target is the next consecutive season's index minus the current one. A season pair only counts with at least 30 matches in both seasons for the top five leagues, and five for other competitions.
- The five features: the current index, total incoming spend, spend on attacking players, average squad age, and the change in average market value.
- How the wrong lag was removed: build one row per club and season first, attach the exact previous year, and only then join competitions. Done in the other order, a club playing three competitions is counted three times before the lag is computed.
- The chronological holdout: earlier seasons train the model, later seasons test it, and the baseline predicts no change at all on those same test rows. A random split would let future seasons leak into training.
- $R^2 = 0.127$ does not mean 12.7% accuracy, and it is not a share of correct predictions. The honest comparison is MAE 0.4856 against the baseline's 0.5114 — about five percent better.
- The Lab uses a separate model fitted on all eligible data for exploration. Its own fit is not an independent test, and moving a budget slider shows a learned association, not the effect of an actual transfer.
- The season comparison also shows goals for, goals against, goal difference and per-match attack and defense ranks derived from real scorelines. Those are separate from the Transfermarkt-based index.
- The defensive-spend follow-up correlates defender fees with the next-season change in goals conceded per match: $r = 0.0040$, $p = 0.427$, $n = 38{,}531$. There is no useful linear signal.

Что показать

1. Open the Project page and read the current r, p, n and MAE aloud, naming the units and the sample behind each one.
2. Explain why a 23/24 → 25/26 pair is invalid even though a naive shift by one row returns it.
3. Change one input in the Lab, return to the baseline, and say what the comparison between the two actually measures.

Короткая речь на английском

> *We define attacking output as recorded goals plus assists per source match. The target is the change in that index in the next consecutive season. We remove incomplete season pairs and construct lagged squad values before joining competitions. Evaluation uses earlier seasons for training and later seasons for testing. The RandomForest has a test MAE of about 0.486, compared with 0.511 for predicting no change: an improvement of roughly five percent. This is a modest signal. The comparison also separates the Transfermarkt attacking index from completed league scores: goals for, goals against, their difference, and per-match attack and defense ranks. A defensive-spend follow-up also has almost zero association with next-season conceded-goal change (r=0.004, p=0.427; n=38,531). The Lab uses a separate all-data model for retrospective exploration. Moving a budget slider changes a model input; it does not establish what a transfer would cause.*

Переход: “The application makes those distinctions visible and lets us inspect the evidence for individual clubs.”

Твой тайминг на защите

Слайд	Блок	Время	Длительность
4. Define the observation before fitting	04 / Method	03:45–05:15	90 с
5. An honest model comparison	05 / Evaluation	05:15–06:30	75 с
7. What the data supports	07 / Findings	09:15–10:15	60 с

Как закрывать экзамен

1. You own slides 4 and 5 (03:45–06:30) and slide 7 (09:15–10:15). Jacopo's 165-second demo sits between them, so watch the clock or you will lose your findings slide.
2. Define the observation before showing the model: club × season × competition, then the target as the difference against the next consecutive season.
3. Say the headline number as a comparison, not an achievement: MAE 0.4856 against a baseline of 0.5114, roughly five percent better.
4. Explain immediately that $R^2 = 0.127$ is not a share of correct predictions. It is the most likely question on your part.
5. If asked about the Lab: it is a separate all-data model for exploration, and its own fit is not an independent test.
6. Use “accompanies” and “is associated with” for every relationship. The word “causes” is more expensive on your part than anywhere else.

7. On xG, answer briefly: the provider field does not exist, we show real shots, and the shots-on-target estimate is an experimental proxy.

Не твоя часть — передай вопрос

1. Инфраструктурные числа не твои: время старта, память, ускорение по индексу и эксперимент роста — часть Атая. Передавай одной фразой.
2. Происхождение данных, загрузка и схема — часть Неды. Ты начинаешь с того, что уже лежит в таблицах.
3. Если спрашивают число, которого нет в карточке, называй файл, а не версию по памяти: docs/findings.md и docs/LAB.md.

Где смотреть в проекте

Файл в проекте	Что в нём смотреть
`pipeline/compute_predictions.py`	Target, пять признаков, лаг рыночной стоимости, хронологический holdout и baseline нулевого изменения — всё в одном файле.
`docs/LAB.md`	Протокол модели и отличие Lab-модели (все данные) от оценочной модели (ранние сезоны → поздние).
`docs/findings.md`	Актуальные r, p, n, R^2 и MAE. Называй числа отсюда, а не по памяти.
`docs/GLOSSARY.md`	Определения Champagne Index, Discipline Index, рангов и граница xG.
`pipeline/feature_correlations.py`	Матрица корреляций признаков и флаг коллинеарности |r| > порога.
`pipeline/shap_analysis.py`	mean |SHAP| на том же held-out сплите, что и R^2 — ответ на «почему не in-sample».
`pipeline/power_ratings.py`	Poisson attack/defense с общим home advantage; в коде прямо написано, что оценка in-sample.
`pipeline/proxy_xg.py`	Прокси по ударам в створ и finishing/defensive luck — почему это не xG.
`pipeline/discipline_effect.py`	Связь красных карточек с пропущенными: +0.341 гола за матч, 1 619 против 16 195 наблюдений.
`tests/test_regressions.py`	Зафиксированные известные дефекты модели и расчётов, чтобы они не вернулись.

Словарь простыми словами

Определения намеренно нетехнические: ими удобно объяснять систему человеку вне курса и отвечать на вопрос, формулировку которого ты не узнал(а).

Container / Docker Compose	A container is one packaged program with everything it needs to run. Compose is the file that starts our four containers in the right order: the database first, then the loader, then the processing pipeline, then the web application.
Relational database / PostgreSQL	A store of tables with fixed columns that can be combined by shared keys. We use PostgreSQL 16. Combining two tables on a shared key is called a join.
Schema	The declared shape of the tables: which columns exist and what type each one holds. It is written in sql/schema.sql.
COPY and index	COPY loads a whole file into a table in one stream. An index is a lookup structure, like the index of a book: it makes selective queries fast and is built after the bulk load.
Pipeline	The processing step between the database and the web page. It reads tables, computes our measures and the model, and writes prepared result tables back.
API / endpoint / JSON	The API is the set of web addresses the page calls. Each address is an endpoint. JSON is the plain-text format of the answer.
Champagne Index	Our own measure of attacking output: recorded goals plus assists, divided by the number of matches the source reports for that club-season. It is a proxy, not an official statistic.
Target and feature	The target is what the model tries to predict — here the change in the index in the next consecutive season. A feature is an input the model uses to predict it.
Chronological holdout	We train the model on earlier seasons and test it on later seasons it has never seen. A random split would mix the future into the training data and flatter the result.
Baseline	The simplest possible prediction, used for comparison. Ours predicts no change at all. A model is only useful if it beats the baseline on the same test rows.
R^2 and MAE	R^2 is the share of variation the model explains, from 0 to 1 — ours is 0.127, which is weak. MAE is the average absolute error in index units — ours is 0.4856 against a baseline of 0.5114, about 5% better. Neither is a percentage of correct answers.
Correlation vs. causation	A correlation (r) measures how two numbers move together, from −1 to +1. It cannot show that one causes the other. Every result in this project is a correlation, including the red-card and spending findings.
Proxy	A measurable stand-in for something we cannot measure directly. Our shots-on-target estimate is a proxy for chance quality; it is not xG, which the data source does not contain.
Poisson rating	A simple statistical model of how many goals each team scores and concedes, fitted to one league-season. It describes that season; it does not forecast new matches.

SHAP	A method that attributes a model's prediction to its individual inputs, so we can report which feature mattered most without relying on the model's internal importance score.

Практические задания

1. Compute (76 + 70) / 38 and explain why 3.84 is not goals per match.
2. Compute (0.511364 − 0.485593) / 0.511364 ≈ 5.04% and state the correct interpretation out loud.
3. Separate three questions: a correlation inside one season, a change in the next season, and the causal effect of buying a player.

Вопросы по направлению

Is Champagne Index an official metric or xG?

Это официальный индекс или xG?

No. It is this project’s proxy: recorded goals plus assists divided by source matches. A goal with an assist contributes twice. We explain the denominator and do not present it as xG or an official rating.

What exactly does the model predict?

Что именно предсказывает модель?

The next consecutive season’s Champagne Index minus the current index. It does not predict a match score, a calibrated league position or the causal return from buying a player.

Why reject gaps between seasons?

Почему нельзя брать просто следующую строку?

A club may leave a competition and return years later. shift(-1) gives the next recorded row, which is not necessarily next year. We explicitly require a one-year gap and adequate coverage in both seasons.

Why use a temporal split?

Зачем хронологический holdout?

A random row split mixes earlier and later seasons and can make historical evaluation overly optimistic. We train on earlier seasons and test on later ones, keeping observations from the same season together.

How much better is the model than a baseline?

Насколько модель лучше baseline?

On the same 5,585 held-out rows, MAE is about 0.4856 versus 0.5114 for predicting no change. That is about a five percent reduction in MAE, a modest gain. R-squared is about 0.127, not a percentage of correct predictions.

Can you conclude that buying attackers improves performance?

Можно ли утверждать причинный эффект?

No. These are retrospective observational data with confounding and source limitations. A slider changes a model input and shows a learned association, not the outcome of a controlled transfer intervention.

Why is the Lab model different from the evaluation model?

Почему две обученные модели?

The evaluation model uses only earlier training data and is scored on held-out later seasons. A separate model is refitted on all eligible data for exploration. Its own training fit is not claimed as independent test performance.

Why did the index-position correlation change after the audit?

Почему изменилась корреляция?

The earlier result mixed a partial 25/26 attacking snapshot with full final standings. We now require adequate comparable coverage. The corrected same-season association is about minus 0.793 over 388 observations.

What is the main result in one sentence?

Главный результат одним предложением?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

Главное ограничение?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

Как в сравнении рассчитываются забитые, пропущенные голы и разница?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Как рассчитываются атакующий и защитный рейтинги?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

Есть ли в данных xG или можем ли мы его оценить?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

Чем атакующий рейтинг отличается от Champagne Index?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

Как вы проверили вратарские показатели по пропущенным?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

Что показывает анализ расходов на защитников?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

Почему defensive-показатели по игрокам отделены от итогов матчей?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

Оценка по ударам — это xG?

No. It multiplies a team’s shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

Что показывают Poisson-ранги атаки и обороны?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project’s held-out prediction check.

Did the added defensive features improve the model?

Улучшили ли защитные признаки модель?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

Красные карточки приводят к большему числу пропущенных?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch $p < 0.0001$). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Какие фразы не использовать

- The model proves that a bigger budget improves performance.
- R^2 is the share of correct predictions.
- The held-out evaluation is a test of the all-data Lab model.

Экзаменационный день

Дата	Формат	Что требуется
12 November 2026 CIDS Lecture 15, pp. 37–39	Checkpoint · 5 minutes	One end-to-end demo, one visible result and the first measured full-run time. A polished interface is not required.
20 / 26 November 2026 CIDS Lecture 15, pp. 37–39	Final defense · 15 minutes + 5 minutes of questions	Every real participant speaks. Suggested split: question and data 3 min, architecture 2, live demo 3, measurements 3, problems 2, contributions 2.

Подготовка машины и материалов

1. Thirty minutes ahead: run `docker compose up -d --build`, then open /api/health and wait for `ready=true` with the loader and pipeline finished.
2. Open four tabs in advance: Overview (Serie A 24/25), the Inter / Milan comparison, the Lab, and the Project page with the numbers.
3. Open the offline study site `defense/site/index.html` and `defense/site/downloads/Presentation.pdf`: both work without a network and without the database.
4. Check that the screenshots in `defense/site/assets/*-desktop.png` are present. They are the fallback if Docker does not come up.
5. Turn off notifications, connect the charger, test screen mirroring and increase the browser font size.
6. Decide in advance who drives the demo and who watches the clock. The timer is built into `defense/site/presentation.html` (N for notes, F for fullscreen).

Как отвечать на вопросы

- Answer in three steps: one short sentence, then the evidence (a file or a number), then the limitation.
- Give every number with its sample and unit: “r = −0.79 across 388 club-seasons, within the same season”, not “the correlation is high”.
- If you do not know, say where the answer lives instead of guessing: “that is in docs/MEASUREMENTS.md, the memory sampling method is described there”.
- If the question belongs to someone else's part, hand it over in one sentence rather than approximating their area.
- Never turn an association into a cause. Say “accompanies” or “is associated with”, never “causes” or “improves”.
- If asked about a weak point, answer it first and directly. An acknowledged limitation is stronger than a defended inaccuracy.

Неудобные вопросы: отвечай прямо и первым

Тема	Вопрос	Ответ и где подтверждение
Licence of the primary source	Is your data openly licensed?	We pinned the source revision and verified headers and checksums, but there is no licence file, so we did not establish an explicit reuse licence. We document it as unresolved and we do not redistribute the raw data. Public downloadability is not a licence. `docs/SOURCES.md, docs/LICENCE_DECISION.md`
Four contributors	Who did what in this project?	Describe only the work you can open and explain on screen. Git identities and role cards are not proof of independent contribution, and the repository says so. `docs/WORK_PACKAGES.md, README.md`
Reproduction on another machine	Did someone else reproduce it?	The run is reproducible on the measured machine, and we also rebuilt it from a clean clone. An independent fresh checkout on another person's laptop has not been recorded yet, so we do not claim it. `docs/FRESH_CHECKOUT.md, docs/MEASUREMENTS.md`
xG and proxies	Is this xG?	No. The source has no provider xG field. We show observed shots and shots on target, and a shots-on-target conversion proxy that we label experimental. The Poisson attack and defense ratings describe the season they are fitted on. `docs/GLOSSARY.md, pipeline/proxy_xg.py, pipeline/power_ratings.py`

> Роль — учебная ответственность. Не заявляй личное авторство или самостоятельную проверку, если этого не было. Покажи ровно те изменения и команды, которые выполнял(а) лично.